

The compact E_6 sigma-MGF on V_{27}

Proof, central selection rule, and verification

Computational proof companion

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Abstract

For simply connected compact E_6 and its minuscule 27-dimensional module, we prove the constant-term formula, the mod-three selection rule, and the initial expansion $1 + h_3 + O_{\geq 6}$. An exact 27-weight computation verifies all Schur coefficients through degree three.

1 Compact Molien–Weyl proof

Expanding the determinant by the Cauchy identity and integrating characters gives

$$\mathcal{S}_{K,V}(\mathbf{t}) = \int_K \prod_a \det(I - t_a \rho(g))^{-1} dg = \sum_{\lambda} \dim(\mathbb{S}_{\lambda} V)^K s_{\lambda}(\mathbf{t}).$$

Direct expansion in the variables also proves the requested identity $[m_{\tau}] \mathcal{S}_{K,V} = \dim(\bigotimes_i \text{Sym}^{\tau_i} V)^K$. For V_{27} all 27 weights form the orbit $W(E_6)\omega_1$, with multiplicity one. Weyl integration, with $|W(E_6)| = 51840$, proves

$$\mathcal{S}_{E_6,27} = \frac{1}{51840} \text{CT}_z \left[D_{E_6}(z) \prod_a \prod_{\mu \in W\omega_1} (1 - t_a z^{\mu})^{-1} \right].$$

2 Selection rule and one-copy series

The generator of the center $Z(E_6) \cong \mathbb{Z}/3$ acts on V_{27} by a primitive cube root. It acts on every degree- d tensor construction by its d th power. Consequently

$$[m_{\tau}] \mathcal{S}_{E_6,27} = 0 \quad \text{unless} \quad 3 \mid |\tau|.$$

The exceptional determinant is the unique symmetric cubic invariant. Thus

$$\boxed{\mathcal{S}_{E_6,27} = 1 + h_3 + O_{\geq 6}}, \quad \boxed{\mathcal{S}_{E_6,27}(t, 0, \dots) = \frac{1}{1 - t^3}}.$$

The module belongs to simply connected E_6 and does not descend to the adjoint quotient E_6/Z_3 .

The all-degree statement $\mathbb{C}[V_{27}]^{E_6} = \mathbb{C}[\det]$ is an external invariant-theory input for the minimal 27-dimensional module. The notebook verifies the cubic invariant and the displayed coefficients but does not independently prove generation in all degrees.

3 Exact verification

The code generates the 72 roots, obtains ω_1 by rationally solving $A^T c = e_1$, and closes its Weyl orbit under simple reflections. The resulting 27 weights define the explicit matrix $M(\theta) = \text{diag}(e^{i\langle \mu, \theta \rangle}) \in U(27)$. The determinant formula and eigenvalue product are evaluated independently.

For coefficients, Jacobi–Trudi produces $\chi_{\mathbb{S}_\lambda V}$. The one-sided Weyl denominator

$$\prod_{\alpha > 0} (1 - z^{-\alpha}) = \sum_{w \in W} \det(w) z^{w\rho - \rho}$$

makes its integral an exact signed weight sum. Orbit membership of $\rho - \mu$ is tested by reflecting it to the dominant chamber. The monomial-basis check gives zero in degrees one and two and

$$([m_{(3)}], [m_{(2,1)}], [m_{(1,1,1)}]) = (1, 1, 1).$$

λ	$\emptyset$	(1)	(2)	(1, 1)	(3)	(2, 1)	(1, 1, 1)
proposed	1	0	0	0	1	0	0
exact Weyl check	1	0	0	0	1	0	0

The orbit contains exactly 27 distinct weights. Unitarity holds to 4.3×10^{-16} , the integrand residual is below 1.8×10^{-15} , and all coefficient checks pass.

Reproducibility. Run `e6_verification.py` with `marimo`.

External invariant-theory source

See T. A. Springer and F. D. Veldkamp, *Octonions, Jordan Algebras and Exceptional Groups*, Springer Monographs in Mathematics (2000), for the Albert determinant and the E_6 structure group.